

MASTER-KIDS · OLYMPIAD PRACTICE SERIES · CLASS 4

Mathematics Olympiad Worksheet 18

Level 1 — Beginner · Data Handling

Class	Class 4 (Primary · Stage 2)	Subject	Mathematics
Level	Level 1 — Beginner	Topic Focus	Data Handling — Bar Graphs, Pictographs & Tally Marks
Questions	20	Time Allowed	30 minutes
Total Marks	20	Worksheet No.	18 of 100
Curriculum	CBSE · ICSE · State Boards (board-agnostic) · NEP 2020-aligned		

About Master-Kids

Master-Kids is India's cradle-to-career learning platform — one platform that grows with a child from Pre-Nursery all the way to college, turning everyday effort into visible progress for parents and a rewarding game of streaks, XP and badges for children. This Olympiad practice series is part of the Master-Kids Primary School programme (Stage 2, Class 1-5). It is board-agnostic — suited to CBSE, ICSE and State Board children alike — and aligned with the NEP 2020 focus on foundational numeracy, science thinking and language.

About This Worksheet

Worksheet 18 teaches the child to READ data the Olympiad way: two bar graphs (find the tallest, the difference, the total), a pictograph where one symbol stands for 2 books (the scale is the trick!), tally marks with crossed bundles of five, and a data table. The final questions test scale reasoning and why graphs matter at all.

Learning Objectives

- Read values, differences and totals from a bar graph.
- Interpret pictographs where one symbol stands for more than one item.
- Count with tally marks using crossed bundles of five.
- Compare and combine values from a data table.
- Order data and reason about pictograph scales.

Instructions

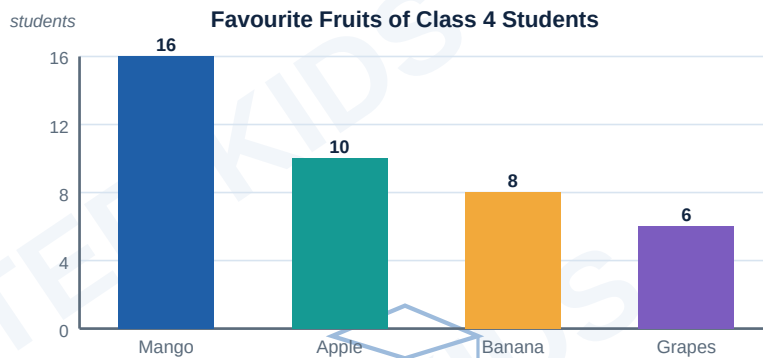
1. This worksheet has 20 multiple-choice questions. Each question has exactly one correct answer.
2. Read every question carefully. Look at any picture, table or figure before choosing.
3. Each correct answer earns +1 mark. There is no negative marking, so attempt every question.
4. Write your chosen option — (A), (B), (C) or (D) — clearly in the answer box or on a separate sheet.
5. Do all rough work in the margin or on rough paper. Do not use a calculator.
6. After finishing, check your answers against the Answer Key, then read the Detailed Solutions to understand the reasoning behind every question — that is where the real learning happens.

Marking scheme: +1 mark for each correct answer · No negative marking · Maximum 20 marks.



SECTION A · Multiple Choice Questions · 20 Questions

Q1. The bar graph shows the favourite fruits of Class 4 students. Which fruit do the MOST students like?



- (A) Mango
(B) Apple
(C) Banana
(D) Grapes

Q2. (Use the same fruit graph.) How many students like **Apple**?

- (A) 8
(B) 10
(C) 12
(D) 16

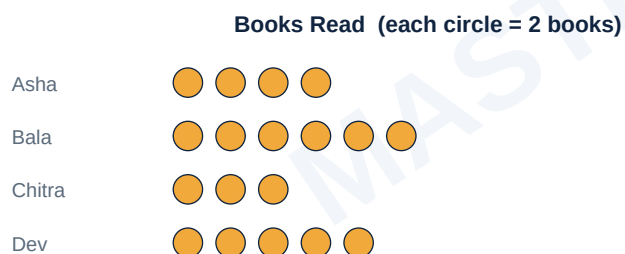
Q3. (Use the same fruit graph.) How many MORE students like Mango than Grapes?

- (A) 6
(B) 8
(C) 10
(D) 12

Q4. (Use the same fruit graph.) How many students were asked in ALL?

- (A) 38
(B) 40
(C) 42
(D) 44

Q5. The pictograph shows books read, where each circle stands for **2 books**. How many books did Bala read?



- (A) 6
(B) 8
(C) 10
(D) 12

Q6. (Same pictograph, 1 circle = 2 books.) How many books did Chitra read?

- (A) 3
(B) 5
(C) 6
(D) 8

Q7. (Same pictograph.) Who read EXACTLY 10 books?

- | | |
|------------|----------|
| (A) Asha | (B) Bala |
| (C) Chitra | (D) Dev |

Q8. (Same pictograph.) How many books did Asha and Chitra read TOGETHER?

- | | |
|--------|--------|
| (A) 7 | (B) 10 |
| (C) 14 | (D) 12 |

Q9. In tally marks, the group |||| with a line across it (5-bar gate) stands for:

- | | |
|-------|--------|
| (A) 4 | (B) 5 |
| (C) 6 | (D) 10 |

Q10. A tally count shows TWO crossed bundles and 3 extra single marks. What number is this?

- | | |
|--------|--------|
| (A) 10 | (B) 11 |
| (C) 13 | (D) 23 |

Q11. Riya counted 17 birds using tally marks. How many CROSSED bundles and single marks did she draw?

- | | |
|-----------------------------|-----------------------------|
| (A) 3 bundles and 2 singles | (B) 2 bundles and 7 singles |
| (C) 3 bundles and 3 singles | (D) 4 bundles and 3 singles |

Q12. A table shows pencils sold: Monday 25, Tuesday 30, Wednesday 20, Thursday 35. On which day were the FEWEST pencils sold?

- | | |
|---------------|--------------|
| (A) Monday | (B) Tuesday |
| (C) Wednesday | (D) Thursday |

Q13. (Same table: Mon 25, Tue 30, Wed 20, Thu 35.) How many pencils were sold in the FOUR days together?

- | | |
|---------|---------|
| (A) 100 | (B) 105 |
| (C) 110 | (D) 115 |

Q14. (Same table.) How many MORE pencils were sold on Thursday than on Monday?

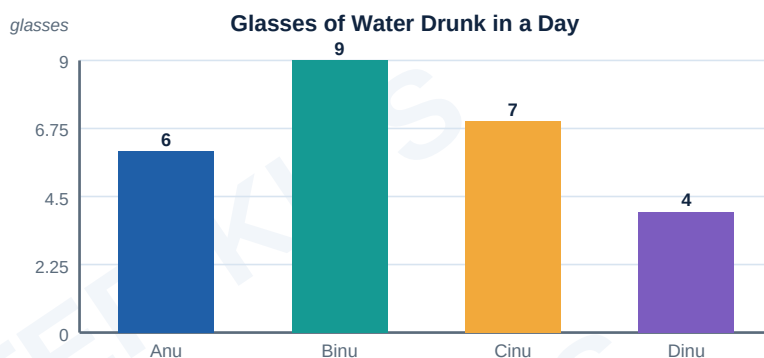
- | | |
|--------|--------|
| (A) 5 | (B) 10 |
| (C) 15 | (D) 20 |

Q15. (Same table.) On which TWO days together were EXACTLY 50 pencils sold?

- | | |
|--------------------------|----------------------------|
| (A) Monday and Tuesday | (B) Tuesday and Wednesday |
| (C) Monday and Wednesday | (D) Wednesday and Thursday |



Q16. The graph shows glasses of water drunk by four children in one day. Who drank the MOST water?



- (A) Anu
(C) Cinu

- (B) Binu
(D) Dinu

Q17. (Same water graph.) How many glasses did Anu and Dinu drink TOGETHER?

- (A) 9
(C) 11

- (B) 10
(D) 13

Q18. (Same water graph.) Arrange the children from MOST to LEAST water drunk.

- (A) Binu, Cinu, Anu, Dinu
(C) Dinu, Anu, Cinu, Binu

- (B) Binu, Anu, Cinu, Dinu
(D) Cinu, Binu, Anu, Dinu

Q19. In a pictograph, one symbol stands for 10 flowers. How many symbols show **40 flowers**?

- (A) 4
(C) 30

- (B) 10
(D) 40

Q20. Why do we use graphs and charts to show data?

- (A) They make numbers harder to read
(C) They use less ink

- (B) They help us see and compare information quickly
(D) They are only for decoration

ANSWER KEY & DETAILED SOLUTIONS

Quick Answer Key

Q	1	2	3	4	5	6	7	8	9	10
Ans	A	B	C	B	D	C	D	C	B	C
Q	11	12	13	14	15	16	17	18	19	20
Ans	A	C	C	B	B	B	B	A	A	B

Step-by-Step Solutions

Q1. Correct answer: (A)

The tallest bar is **Mango** (16 students).



Q2. Correct answer: (B)

The Apple bar reaches **10** on the scale.

Q3. Correct answer: (C)

Mango = 16, Grapes = 6. Difference = $16 - 6 = 10$ students.

Q4. Correct answer: (B)

Add all bars: $16 + 10 + 8 + 6 = 40$ students.

Q5. Correct answer: (D)

Bala has 6 circles, and each circle = 2 books: $6 \times 2 = 12$ books.

Q6. Correct answer: (C)

Chitra has 3 circles: $3 \times 2 = 6$ books.

Q7. Correct answer: (D)

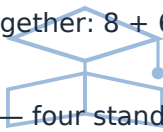
10 books = $10 \div 2 = 5$ circles. **Dev** has 5 circles.

Q8. Correct answer: (C)

Asha = $4 \times 2 = 8$ and Chitra = $3 \times 2 = 6$. Together: $8 + 6 = 14$ books.

Q9. Correct answer: (B)

A crossed bundle of tally marks stands for **5** — four standing lines and one across.



Q10. Correct answer: (C)

Two bundles = $2 \times 5 = 10$, plus 3 singles = **13**.

Q11. Correct answer: (A)

$17 = 5 + 5 + 5 + 2 \rightarrow$ **3 crossed bundles and 2 single marks**.

Q12. Correct answer: (C)

Compare: 25, 30, **20**, 35. The smallest is Wednesday with 20 pencils.

Q13. Correct answer: (C)

$25 + 30 + 20 + 35 = 110$ pencils.

Q14. Correct answer: (B)

$35 - 25 = 10$ more pencils.

Q15. Correct answer: (B)

Tue + Wed = $30 + 20 = 50$. (Mon+Tue=55, Mon+Wed=45, Wed+Thu=55.)

Q16. Correct answer: (B)

The tallest bar is **Binu** with 9 glasses.

Q17. Correct answer: (B)

Anu 6 + Dinu 4 = **10 glasses**.

Q18. Correct answer: (A)

$9 > 7 > 6 > 4 \rightarrow$ **Binu (9), Cinu (7), Anu (6), Dinu (4).**

Q19. Correct answer: (A)

$40 \div 10 =$ **4 symbols**.

Q20. Correct answer: (B)

Graphs turn numbers into pictures so we can **see and compare information quickly** — the tallest bar or longest row jumps out at once.

